

MedGov-AI: An MCP-Based Agentic Orchestrator for Clinical Medical Image Analysis and Structured Report Generation

Abstract—Clinical AI tools for medical imaging are increasingly capable but remain largely isolated from one another, requiring custom integration work that slows adoption and creates fragmented, difficult-to-maintain deployments. This paper presents MedGov-AI, an agentic AI system that addresses the integration and orchestration problem directly by using the Model Context Protocol (MCP) as a standardized interoperability layer. Rather than hardcoding workflow logic, MedGov-AI uses a large language model as an autonomous orchestrator that discovers and invokes clinical tools at runtime through a set of MCP servers covering DICOM parsing, CT segmentation (MONAI), whole-slide image cell segmentation (Cellpose), radiology report generation (RadLex), and digital pathology (iPath). Domain-specific workflow governance is provided through skill files, plain-text protocol documents that encode step-by-step clinical sequences and guard conditions without retraining the model. The system supports both interactive and event-driven operation: in workspace monitoring mode, files deposited into a configured directory trigger autonomous agent execution without any user input. Two end-to-end use cases were evaluated: DICOM-based CT analysis with structured report generation, and whole-slide pathology tumor detection with automated cell segmentation and cross-workspace file routing. Results show that API-hosted large language models complete multi-step clinical workflows reliably within a bounded iteration budget, while locally-hosted 7-8B parameter models do not, revealing a current tension between privacy and reliability in clinical deployment. Skill-guided execution prevented domain-incorrect tool selection in the pathology use case, confirming that skills are necessary at decision forks where plausible but clinically incorrect options exist. Assisted and autonomous modes produced identical tool sequences, validating unattended autonomous operation. Six MCP servers were integrated without any agent-side connector code: each server wraps its underlying tool in the MCP protocol, and the agent discovers and invokes all of them through the same standard interface, supporting the argument that MCP can reduce the integration overhead that currently limits clinical AI adoption.

Index Terms—Agentic AI, Medical Imaging, MCP, MONAI, Clinical Workflow Automation

I. BACKGROUND

A. AI in Clinical Practice

Healthcare systems worldwide face increasing demand with limited resources. The volume of medical imaging studies continues to grow while the number of specialists remains constrained, creating diagnostic backlogs that directly affect patient outcomes. Automating routine analysis tasks is therefore not merely a convenience but a structural necessity. Greater automation brings greater consistency, reduces the workload on healthcare professionals, accelerates diagnosis,

and allows clinicians to dedicate more time to complex cases and direct patient care. Two questions frame the motivation for this work:

What AI tools are deployed in clinical settings and what do they do? Several AI tools are currently used across clinical disciplines, with the notable characteristic that they typically operate in isolation from one another, with no unified integration layer. In radiology, AI assists with worklist prioritization and critical finding alerts, where the system detects life-threatening conditions such as stroke, pulmonary embolism, or intracranial hemorrhage and immediately notifies the clinical team. Mass General Brigham (United States) has deployed AI-driven radiology worklist prioritization to reduce time-to-read for urgent studies [1]. The NHS (United Kingdom) deployed Annalise.ai across more than 40 hospital trusts for chest X-ray triage, allowing radiologists to focus on flagged abnormal cases first [2]. In screening programs, AI acts as a first filter, flagging high-risk cases for specialist review and allowing programs to scale without proportionally increasing the number of specialists required. Google Health conducted a published clinical trial of an AI system for diabetic retinopathy screening in Thailand, demonstrating that AI-assisted grading matched or exceeded specialist performance in a real clinical setting [3]. Mayo Clinic (United States) has integrated AI into ECG interpretation and cardiac imaging workflows as a decision support layer for cardiologists [4]. In pathology, whole slide image analysis is used for tumour grading and cell counting, and AI-generated preliminary reports from imaging findings are now in active clinical use. For segmentation and detection specifically, FDA-cleared tools such as Viz.ai [5] and Aidoc [6] are deployed for CT-based triage; Transpara [7] for mammography screening; and IDx-DR [8] for diabetic retinopathy grading from fundus images.

What are the benefits and what are the risks? The benefits of AI in clinical workflows are directly tied to the limitations of the current manual process. Faster analysis is the most immediate gain: AI systems can process and flag findings in seconds, reducing the time between acquisition and clinical decision. Consistency is a second key advantage, as AI models apply the same criteria uniformly across every case, unlike human specialists who are subject to fatigue, experience, and workload. Reducing routine reads and preliminary reporting also frees specialists for cases that genuinely require expert judgement. Taken together, these properties position AI not as a replacement for clinicians but as a force

multiplier. The risks are equally significant. Bias is a primary concern: models trained on data from specific demographics or equipment may underperform on underrepresented groups or institutions using different hardware. Hallucination is critical in language models used for report generation, where confident but clinically incorrect statements carry direct patient risk [9]. The black-box nature of deep learning raises accountability concerns, as clinicians and regulators often cannot explain why a model reached a particular conclusion [10]. From a regulatory standpoint, no unified framework governs how AI tools in healthcare should be validated, updated, or audited in production; Jia et al. [11] further showed that regulatory approval at development time does not cover new hazards that emerge only after deployment. Finally, and most relevant to this work, no established standards govern how agentic AI systems should operate in clinical environments, including how they communicate with existing tools, when they escalate to a human, and how their decisions are logged for audit [12]. The gap between technical capability and clinical deployment is substantial: Cozzolino et al. [13] found that fewer than 4% of AI studies had been evaluated in real clinical settings, and Kondylakis et al. [10] reported that over 80% of radiologists had not integrated AI into their daily practice.

Despite the progress of individual AI tools, no standard integration layer exists to orchestrate them together within a clinical workflow. Each institution must develop its own connectors and its own standards, which leads to fragmentation, lack of interoperability, slow adoption, and deployments that are expensive and difficult to maintain. The Model Context Protocol (MCP)¹ is a promising solution to this problem, providing a standardized way for agents to discover and call tools without custom integration. This work explores the application of MCP as the integration layer for an agentic AI system that orchestrates multiple clinical imaging tools in a real-world workflow.

B. Agentic AI in Healthcare

Agentic AI² refers to systems that can autonomously set goals, plan multi-step actions, and use tools to achieve those goals [14], [15]. Unlike traditional AI models that operate in a single step or require human input at every stage, agentic systems can reason through complex tasks, make decisions about which tools to use, and adapt their behavior based on feedback. The application of agentic AI in healthcare is still in an early stage, with systematic reviews identifying very few systems that have been evaluated in real clinical settings [14], [16]. This makes them particularly well-suited

for clinical imaging workflows, which are inherently sequential and often require the integration of multiple tools and data sources. For example, a typical workflow for analyzing a CT scan might involve first running a segmentation model to identify regions of interest, then using a classification model to characterize those regions, and finally generating a structured report summarizing the findings. An agentic AI system can orchestrate this entire process autonomously, calling each tool in the correct sequence and synthesizing the results into a coherent output.

Three core properties characterize agentic systems: goal persistence (the system does not stop until the required deliverables are met), tool use (it can act on external systems, not just generate text), and iterative reasoning (each result informs the next decision). We evaluated several existing agentic frameworks, including LangChain [17], CrewAI [18], and AutoGen [19], but opted to build a custom agentic system. The primary reason was the need for tight integration with MCP: existing frameworks do not natively support MCP as a tool protocol, which would have required additional adapter layers and reintroduced the fragmentation problem we were trying to solve. Building a custom loop also allowed us to enforce specific behaviors required in a clinical context, such as a completion contract under which the agent cannot declare success while required deliverables remain unmet, and a debug mode in which the user must approve each tool call before execution.

Contrasting with simpler alternatives clarifies why this design was necessary. A plain chatbot produces one response per prompt and cannot maintain state or call tools. A hardcoded pipeline can call tools but relies on the developer to anticipate every possible execution path, making it brittle in the face of real-world variability. A standalone ML model provides predictions but has no language interface and cannot adapt its behavior based on context. We found that only a true agentic system could handle the complexity and variability of real clinical workflows, which often require conditional logic, error handling, and dynamic tool selection.

Our own system began as precisely this kind of hardcoded pipeline: in the initial version, the LLM was never consulted, and all sequencing decisions were made by explicit conditional logic in the orchestrator code. This approach proved limiting, as it could not adapt to new information without significant reprogramming, which motivated the transition to a fully agentic design.

This transition is not without its challenges. Reliable agentic behavior requires careful prompt engineering: the agent must be guided to select tools in the correct order, avoid premature declarations of success, and handle ambiguous or incomplete results gracefully. These challenges are discussed in detail in later sections. However, the fundamental advantage over any hardcoded alternative is scalability: supporting a new clinical domain does not require rewriting workflow logic but only adding a new MCP server that exposes the relevant tools, or a skill file that encodes the domain-specific protocol, and the agent discovers and incorporates both at runtime without code

¹MCP is an open protocol that defines a standardized client-server interface through which an AI agent discovers and invokes external tools. Each MCP server declares its available tools at startup with names, input schemas, and natural language descriptions; the agent selects and calls them by name at runtime without any prior knowledge of the server’s implementation.

²An agentic AI system is one that pursues a goal through iterative reasoning and tool use rather than producing a single response. At each step the model decides what action to take next, executes it via an external tool or API, observes the result, and continues until the goal is reached or the iteration budget is exhausted.

changes. This also opens a much wider architectural design space. Rather than a single fixed agent, one could configure specialist agents for distinct clinical tasks, use prompt-based behavioral switches to change the agent’s operating mode, or compose multiple MCP servers covering entirely different domains. None of these extensions require touching the orchestration code, which is simply not true of any hardcoded pipeline.

When testing the system we explored two LLM backend approaches: a locally-hosted model and a cloud API. The local model was Ollama’s deepseek-r1:7b (4.7 GB), with llama3.1:8b (4.9 GB) also evaluated, both small enough to run on consumer hardware and allowing for complete data privacy since all processing happens locally. The API-based models tested were Gemini 2.0 Flash, Gemini 2.5 Flash, and Gemini 3.0, all significantly larger and more capable models hosted in the cloud. The local approach offers greater control and privacy, though privacy considerations were not the primary focus of this work and are left for future investigation. The local approach proved to struggle with the demands of multi-step clinical workflows. We observed consistent failures in tool calling: the model frequently failed to select the correct tool for the given context and could not reliably chain multiple calls to reach a goal. Addressing this consumed considerable development time across prompt engineering, workflow simplification, logging, and targeted adjustments. None of these fully resolved the problem, as the root cause was model scale rather than configuration: smaller models have fewer parameters and correspondingly less capacity for the multi-step contextual reasoning required to chain tool calls reliably across a clinical workflow. Beyond raw capability, API-hosted models also represent a more accessible deployment path for clinical institutions. Acquiring and maintaining the GPU infrastructure required to run large local models is a significant investment that many healthcare organizations cannot justify, whereas access to a capable cloud model requires only an API subscription, a constraint independently confirmed across the field [20]. Switching to the API-hosted model resolved all of these issues immediately. The API approach introduces trade-offs of its own: latency, potential data security concerns in privacy-sensitive deployments, and a dependency on third-party service availability that is outside the institution’s control. If the API experiences downtime or degraded performance, the entire orchestration layer is affected, which is a meaningful reliability concern in clinical settings. Despite these limitations, for the complex reasoning required by clinical imaging workflows it proved to be the only reliably viable option at the model sizes currently available for local inference. The system architecture supports both backends, allowing deployment to be tailored to the constraints of a given clinical environment.

C. MCP as a Standardized Tool Integration Layer

The conventional approach to agent-tool integration is the custom REST adapter: for each tool the agent must call, a custom connector is written that handles authentication, request serialization, response parsing, and error mapping. This

approach does not scale. Each new tool adds a new adapter; changes to a tool API break the corresponding adapter; and the agent itself must encode assumptions about every tool it might call.

The Model Context Protocol (MCP) addresses this directly. MCP defines a client-server protocol in which each tool server declares its available tools at startup, specifying names, input schemas, and descriptions. The agent discovers tools dynamically at runtime and invokes them by name without requiring any prior knowledge of the server’s internal implementation. This decouples the agent from individual tool APIs: adding a new tool server requires no changes to the agent, only a new MCP-compliant server that exposes the relevant capabilities.

Singh et al. (2025) survey MCP’s architecture in detail, positioning it as the solution to fragmented API integrations for agentic AI systems [21]. The survey also identifies MCP sampling, a mechanism for LLM-to-LLM communication within the MCP protocol, as a path toward sub-agent specialization, in which a coordinating agent can delegate to specialist agents exposed as MCP servers. Singh et al. note that healthcare is a domain where these properties are especially relevant, but the survey discusses clinical applications only conceptually; no implemented system is presented. A preliminary working integration [22] demonstrated 95% correct tool selection from natural language queries and confirmed MCP’s role as a secure interoperability layer. Conceptually, MCP has been proposed as a foundational layer enabling context-aware, scalable integration of AI into clinical workflows [23], but production-grade clinical deployments remain scarce.

D. Skills as Clinical Capability Packages

Tool access via MCP determines what an agent can do, it does not determine what the agent should do in a given clinical domain. A segmentation server, a terminology server, and a report template server can all be exposed as MCP tools, but the correct sequencing of those tools for a CT lung evaluation versus a pathology workflow requires domain knowledge that is separate from both the agent’s general reasoning capability and the tool’s own logic.

Skills³ address this layer. Xu and Yan (2026) formalize agent skills as composable, on-demand capability packages that provide domain-specific behavioral guidance to an LLM without retraining it [24]. The key architectural insight is that skills and MCP are complementary rather than competing: skills define what to do in a given domain, while MCP exposes the tools with which to do it. Xu and Yan do not address healthcare-specific skill design, the governance question of who maintains skill files when clinical protocols change, or how skills compose with MCP in a working system, gaps that this work directly addresses.

³In this context, a skill is a plain-text file that encodes domain-specific behavioral guidance for the agent - step-by-step workflow protocols, guard conditions, sequencing constraints, or bodies of domain knowledge such as regulatory standards or API usage patterns. Skills are not executable code; they are instructions that the agent reads and follows using its existing tools. The agent loads a skill on demand rather than at startup, keeping the system prompt minimal until a specific domain is required.

E. Positioning of This Work

Existing work addresses components of this problem in isolation. Singh et al. [21] and [22] evaluate MCP as an integration protocol but do not demonstrate it in real clinical imaging workflows. Conceptual frameworks [23] identify clinical information fragmentation and single-purpose tool limitations as core obstacles to clinical AI adoption but stop short of a working implementation. Work on medical imaging AI [20] focuses on model accuracy rather than orchestration, and confirms that deploying capable models locally in clinical institutions is a structural constraint rather than an isolated engineering problem. The skills approach for agentic systems has been formalized conceptually [24] but not applied to clinical workflow governance or evaluated against a no-skill baseline.

MedGov-AI addresses the integration and orchestration layer directly: it combines DICOM parsing, segmentation inference, and structured report generation under a single agentic orchestrator using MCP as the interoperability layer and skill files to encode domain-specific workflow protocols. Both the MCP integration and the skill-based workflow governance are evaluated in end-to-end clinical workflows, providing empirical evidence for properties that prior work has only argued conceptually.

II. METHODS

A. System Architecture Overview

MedGov-AI is organized into three layers: a React frontend, a FastAPI orchestrator, and a set of MCP tool servers. Each layer has a clearly defined responsibility, and the boundaries between them are designed so that clinical domain logic lives neither in the interface nor in the transport layer but entirely in the orchestrator, where it can be reasoned about, logged, and audited.

Frontend: The client layer is a React 18 application that provides the clinical user interface, communicating with the orchestrator exclusively via a REST and SSE API. SSE enables real-time streaming of task progress and agent responses to the user without polling.

Orchestrator: The orchestrator is a FastAPI application and is the system’s single point of control: it exposes the REST and SSE API consumed by the frontend, owns all session and task state, manages connections to every MCP server, and hosts the three internal services that drive the system’s behaviour: the `AgenticAgent`, the `task_runner`, and the `watcher_service`.

The `AgenticAgent` is the sole decision-maker in the system. A deliberate choice was made to use a single agent rather than a hierarchy of sub-agents, for two reasons: simplicity and auditability. A single agent is easier to reason about, debug, and extend, and every tool call, its inputs, and its result are recorded in a single session trace with no hidden delegation. The agent runs an iterative LLM loop using either the Gemini API or a locally-hosted Ollama model as its reasoning backend, and executes multi-turn function calling to

invoke MCP tools in sequence. It maintains a session context across turns so that information produced by one tool call is available to the next without requiring the user to restate it.

The `task_runner` manages long-running operations that cannot complete within a synchronous HTTP request. Operations such as MONAI inference or Cellpose segmentation are submitted to a background thread pool and tracked through a state machine with four states: queued, running, completed, and failed. A semaphore limits concurrent GPU inference to a single job at a time, preventing out-of-memory crashes on constrained hardware. The task runner also captures stderr output from inference processes and, on failure, passes the log to the LLM for a plain-language error explanation before surfacing it to the user. Task state is persisted in SQLite so that progress survives backend restarts, and the frontend is notified of state transitions via SSE.

The `watcher_service` monitors registered filesystem directories using periodic polling and triggers the agent automatically when a new file is deposited. This enables event-driven clinical workflows in which files arriving from a PACS⁴ system or a scanner initiate the full analysis pipeline without any user action. No changes to upstream systems are required; the watcher integrates at the filesystem level and streams its activity log to the frontend via a dedicated SSE⁶ channel.

All persistent state is stored in a single SQLite database with WAL journaling. The schema covers users, sessions, conversation history, uploaded files, background task status, watched directories, and per-user tool settings.

LLM Backends: The orchestrator supports two LLM backends, selectable at deployment time without modifying any agent logic. The Gemini API provides a large cloud-hosted model capable of reliable multi-step function calling in complex clinical workflows, at the cost of external data transmission and API dependency. Ollama provides a local inference alternative using smaller open-weight models running entirely on the deployment host with no external data transmission, which is relevant in privacy-sensitive clinical environments, but proved insufficient for reliable tool calling at the 7–8B parameter scale currently available for local inference. Both backends implement the same internal interface.

MCP Server Layer: The tool layer consists of six MCP servers, each running as an independent process in its own isolated Python virtual environment. Isolation at the process level is a deliberate design choice: it prevents dependency conflicts between servers with incompatible requirements (most critically between the GPU-enabled PyTorch installation required by `mcp-monai` and the CPU-only installation used elsewhere, which cannot coexist in a single environment). Tool names are prefixed with the server name (e.g., `monai.run_inference`,

⁴Picture Archiving and Communication System (PACS): the institutional system used in clinical radiology to store, retrieve, and distribute medical images, typically in DICOM⁵ format.

⁶Server-Sent Events (SSE): a unidirectional HTTP streaming mechanism through which the server pushes events to connected browser clients in real time, without the client needing to poll.

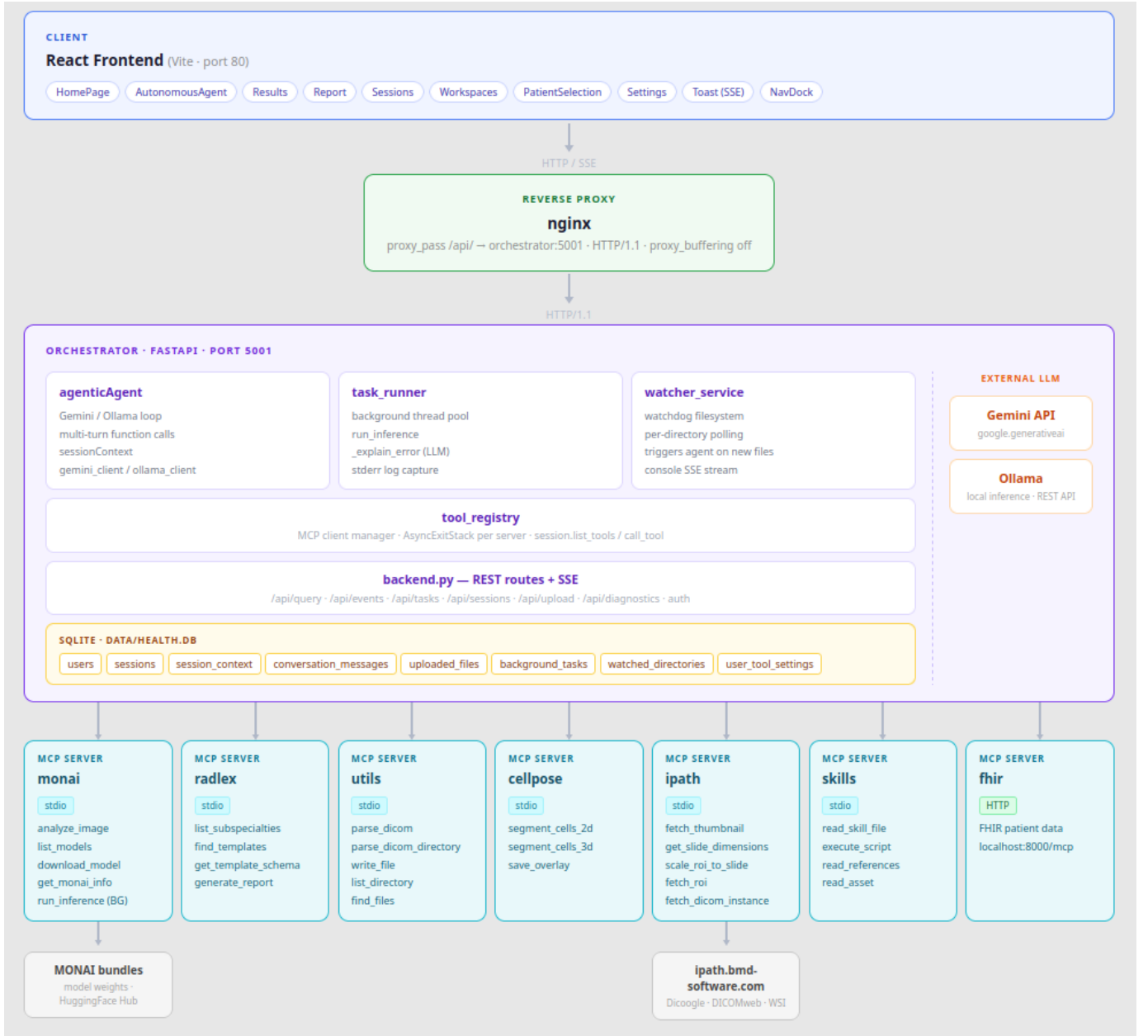


Fig. 1. MedGov-AI system architecture.

`radlex.generate_report`), providing unambiguous namespacing across servers. All servers communicate over stdio transport. The available servers and their clinical functions are listed in Table I.

This architecture means that extending the system with a new clinical capability requires only writing a new MCP-compliant server and registering it in the configuration file. The agent, the orchestrator, and every existing server remain unchanged. This is the property that makes MCP a viable interoperability layer rather than yet another custom integration approach.

B. MCP Server Layer

Each MCP server encapsulates a distinct clinical capability and exposes it as a set of named, schema-defined tools. The agent has no hardcoded knowledge of any server’s internals. It discovers available tools at startup and selects among them based on context.

Mcp-monai is the medical image inference server, built on the MONAI framework [25] with PyTorch as the underlying deep learning backend. It detects the imaging modality and body part automatically from pixel statistics and file metadata, returning a ranked list of recommended models. The model catalogue can be filtered by modality, body part, or category,

TABLE I
MCP SERVERS AND CLINICAL FUNCTIONS

Server	Transport	Clinical Function
mcp-monai	stdio	Medical image segmentation and detection
mcp-radlex	stdio	RadLex terminology and report templates
mcp-utils	stdio	DICOM parsing and filesystem operations
mcp-skills	stdio	Skill workflow loading and execution
mcp-ipath	stdio	iPath telepathology integration
mcp-cellpose	stdio	2D/3D cell segmentation (Cellpose)

and bundles are retrieved from HuggingFace Hub and cached locally on demand. Inference uses a sliding window strategy to handle the variable dimensions of 3D medical volumes. Bundled models cover spleen segmentation on CT, pancreas and tumour segmentation on CT, lung nodule detection on CT, brain tumour segmentation on multi-channel MRI, whole-body CT segmentation across 104 anatomical structures, and kidney segmentation using the UNeST architecture.

Mcp-radlex provides structured radiology reporting through integration with the RadLex terminology standard. The agent can browse available radiological subspecialties, search the template catalogue by subspecialty or keyword. When populating a report from raw inference results, the server uses MCP sampling to delegate the mapping of segmentation outputs to template fields back to the host LLM, which also generates the findings narrative, impression, and recommendations. This closes the loop between inference output and clinical documentation: the agent receives segmentation results, interprets them, and produces a report conforming to a standard radiological schema rather than free text.

Mcp-utils provides general-purpose filesystem and DICOM operations that underpin every clinical workflow. It extracts the full metadata of a DICOM file, including modality, body part, patient identifiers, acquisition parameters, and pixel data statistics, and can apply the same extraction across an entire directory. On the filesystem side, the server supports reading, writing, moving, copying, and deleting files, listing directory contents, and searching by pattern.

Mcp-cellpose provides cell segmentation for pathology workflows using the Cellpose v4 universal model (cpsam). Cellpose v3 exposed multiple task-specific models (cyto3, cyto2, nuclei, each approximately 80 MB) selected by a `model_type` argument. Cellpose v4 replaces all of these with a single SAM-based universal model (cpsam, approximately 2.5 GB) that was evaluated on both versions during development. Testing on the same pathology ROI image showed v3 (cyto3) detecting 1,100 cells against v4 (cpsam) detecting 1,703 cells, a 55% increase in detected count on identical input. v4 was selected as the deployment model. However, cpsam cannot distinguish cytoplasm from nuclei on standard RGB images without separate fluorescence channels, and the `model_type` argument accepted by the v3 API is silently ignored in v4. The server supports both 2D and 3D inputs and composites the predicted cell mask onto the original image for visual inspection. GPU availability is detected at

startup with automatic CPU fallback, though CPU execution is only practical for small test images (see Section III).

Mcp-ipath integrates with the iPath telepathology platform, which aggregates whole slide images (WSI) and DICOM series via Dicoogle and DICOMweb. It supports thumbnail retrieval, resolution-level coordinate mapping, and region-of-interest extraction at arbitrary resolution levels without downloading the full WSI, as well as access to individual DICOM frames and series.

Mcp-skills exposes the skill system to the agent as callable tools, allowing it to load a skill’s entry point and its associated reference documents and assets, and to execute predefined scripts associated with a skill. The skills system is described in detail in Section II-C.

C. Skills System

The skills system provides a mechanism for encoding domain-specific behavioral guidance that the agent loads on demand. A skill body can take different forms depending on the domain: a step-by-step workflow protocol with sequencing constraints and cross-server coordination, or a body of domain knowledge such as regulatory standards, clinical report templates, and API usage patterns. What these forms share is that the guidance cannot be derived from MCP tool schemas alone and is too domain-specific or voluminous to embed permanently in the system prompt. Embedding it permanently would increase token consumption on every request, including those that do not require that domain, and would increase the risk of the LLM overlooking or misapplying instructions buried in a large prompt, a well-documented failure mode in prompt engineering [26]. Skills address both problems by keeping the system prompt minimal and loading domain detail only when the agent determines it is needed for the current task.

1) *Structure*: A skill separates awareness from detail. Every available skill is surfaced to the agent at startup through a short name and natural language description kept permanently in the system prompt, while the full skill body remains unloaded until needed. When a request arrives, the agent reasons over these descriptions to identify the applicable skill, then loads its body on demand as operating instructions for that task. This enables progressive disclosure: the context window carries only a lightweight index at rest, expanding to include full domain detail only for the domain currently in use.

A skill body can itself be hierarchical, containing an entry-level document that routes to more specific sub-documents,

reference material, templates, and scripts, each loaded independently as the task requires. This allows a single skill to cover a broad domain without forcing the agent to load everything at once.

2) *Implemented Skills*: Four skills are currently deployed in the system. The **dicom-analysis** skill encodes the tool sequence and guard conditions for DICOM analysis ending in a radiology report; the **wsi-tumor-detection** skill encodes the visual inspection and cell-counting workflow for whole-slide pathology images. The **clinical-reports** and **pydicom** skills are publicly available community skills [?]: **clinical-reports** encodes regulatory standards and structured templates for clinical documentation, and **pydicom** provides API guidance for reading, writing, anonymizing, and manipulating DICOM files programmatically. The imaging skills illustrate skills as workflow protocols; the community skills illustrate skills as domain knowledge packages, showing that the same mechanism covers both forms of guidance and that externally maintained skills can be adopted without modification.

3) *Governance*: A key property of this design is that skill files are plain text and can be updated without touching the agent, the MCP servers, or any other part of the codebase. When a workflow step changes (a different tool call order, a new guard condition, a revised output format) the relevant reference document is edited and the agent adopts the new behaviour immediately on the next session, with no software deployment required. This separates workflow maintenance from software development, which is relevant in clinical settings where the people who understand the correct procedure are not necessarily the people who maintain the codebase.

D. Agentic Execution Loop

The agent’s operation is governed by an iterative execution loop that drives the system from a natural language goal to a set of completed deliverables, calling MCP tools at each step based on the LLM’s reasoning about what to do next.

1) *System Prompt and Session Initialisation*: At the start of each session the agent receives a system prompt that establishes its role, its available tools, and its behavioural rules. The system prompt includes the list of available skills as a plain bulleted list of names and descriptions, injected at startup after dynamic discovery, so the agent knows what clinical domains it can operate in before any user message arrives. It also includes explicit rules for tool usage: when to read a skill file before acting, when to queue a long-running operation to the background task runner rather than calling it inline, how to handle DICOM series directories versus individual files, and how to avoid repeating failed tool calls. Two variants of the system prompt exist: one for autonomous mode, which omits confirmation-related guidance, and one for assisted mode, which includes it.

2) *Iterative Reasoning and Tool Dispatch*: Each iteration of the loop follows the same structure. The agent receives a prompt describing the current goal, any available files, and the results of all prior tool calls in the session. On the first iteration the prompt is minimal: goal, data, and file context. On

subsequent iterations the prompt also includes the execution history, which the agent evaluates before deciding its next action. The LLM responds with either one or more function calls, or a text response indicating that the goal is complete or that more information is needed. Multiple tool calls can be returned in a single LLM response and are executed sequentially within the same iteration. All results are collected and returned to the LLM in a single message before the next iteration begins. Function calls are parsed and dispatched to the appropriate MCP server via the tool registry, and the result is appended to the execution history.

A repetition guard addresses a failure mode observed during development in which the agent would attempt the same failing tool call repeatedly rather than trying an alternative approach, consuming iterations without making progress. The guard inspects the execution history before each tool call: if the same tool has failed twice in a row, the third attempt is blocked and an error message is injected into the history instructing the agent to try a different tool or approach.

3) *Built-in Tools*: Alongside the MCP tools discovered at startup, the agent has access to a set of built-in tools that are handled directly by the orchestrator rather than by any MCP server. `goal_achieved` signals that the task is complete and carries a summary of the result; calling it is the only way for the agent to exit the loop successfully. `need_more_info` allows the agent to pause execution and ask the user a clarifying question when the goal is ambiguous or required inputs are missing. `queue_task` submits a long-running operation to the background task runner and returns immediately, allowing the agent to respond to the user without blocking on inference. `list_tasks` retrieves the current status of queued or running background tasks, allowing the agent to check whether a previously submitted inference job has completed before proceeding. These tools give the agent explicit control over its own lifecycle and over the background task queue without exposing those concerns as MCP tool servers.

4) *Operating Modes*: The system supports two operating modes that can be selected per session. In **autonomous mode**, the agent executes tool calls without interruption from goal receipt to completion. This is the intended mode for event-driven workflows triggered by the workspace watcher, where no user is present to intervene. In **assisted mode** (referred to in the codebase as debug mode), the loop pauses before each tool call and returns a `confirmation_required` response to the frontend, which presents the proposed tool name and arguments to the user for approval or rejection before execution resumes.

E. Background Task Queue

Inference operations such as MONAI CT segmentation and Cellpose cell counting are long-running by nature, taking anywhere from seconds to several minutes depending on hardware and input size. Blocking the agent or the HTTP request while waiting for these operations to complete would make the system unresponsive to other users and interactions.

To avoid this, all inference calls are offloaded to a background thread pool: when the agent calls a tool that queues inference, the tool returns immediately with a task identifier, and the actual computation runs asynchronously.

Each task follows a simple state machine: `queued` when submitted, `running` once a thread picks it up, and `completed` or `failed` on exit. Task state is persisted in a SQLite database so that results survive server restarts and can be retrieved by any session. A semaphore limits concurrent inference to a single job at a time, preventing simultaneous runs from exhausting GPU memory. When a job finishes, a Server-Sent Events (SSE) notification is pushed to the frontend, which updates the results tab without requiring the user to poll or refresh. This design allows multiple users to submit jobs independently: each submission is queued and processed in order, with results delivered to the corresponding session on completion.

F. Workspace Monitoring and Event-Driven Triggering

Not all clinical workflows are initiated by a user interacting with a chat interface. In many settings, imaging files are deposited into shared directories as part of existing institutional processes, with no operator present to trigger analysis manually. The workspace monitoring system addresses this by turning filesystem events into agent triggers, allowing MedGov-AI to process incoming files automatically without requiring modification to upstream workflows.

1) *Architecture*: Each workspace is a named, registered directory stored in the SQLite database alongside a configurable prompt that defines what the agent should do when a file arrives. At startup, the orchestrator resumes watching all previously registered workspaces. The `watcher_service` uses a polling observer to monitor each registered path recursively, bridging filesystem events into the orchestrator’s async event loop via an internal queue.

When a new file is detected, the service does not trigger the agent immediately. Instead it starts a three-second debounce window: if additional files arrive within that window, they are batched together. Only after three seconds of inactivity does the service dispatch all accumulated files to the agent in a single invocation. This prevents a burst of files, such as a DICOM series arriving slice by slice, from spawning multiple redundant agent executions.

2) *Trigger Flow*: When the debounce window closes, the watcher copies the incoming files from the watched directory into the corresponding workspace directory and constructs a goal prompt that includes the workspace name, the list of received files, and the preconfigured prompt for that workspace. It then invokes the agent in autonomous mode, since no user is present to confirm tool calls. The agent proceeds through its standard execution loop: reading the appropriate skill file (if needed), parsing the files, running the required analysis, and producing the configured outputs. The entire activity log, including file detection events, agent actions, and results, is streamed to the frontend via SSE under the workspace’s dedicated console channel, giving an operator

real-time visibility even when the workflow was triggered without user interaction.

G. Use Cases

Two clinical workflows were selected to evaluate the system. Together they cover the two imaging modalities the system targets, radiology and pathology, and exercise the full tool stack: DICOM parsing, segmentation inference, report generation, and whole slide image analysis. Both workflows can be initiated interactively through the chat interface or triggered automatically when files arrive in a monitored directory.

1) Radiology: DICOM Analysis and Report Generation:

A clinician or automated trigger provides a DICOM file or series. The system identifies the study type, runs the appropriate segmentation model, and produces a structured preliminary radiology report matched to the modality and anatomy. Success is defined as a correctly identified study type, a completed segmentation, and a filled report template with findings mapped to the appropriate fields.

2) *Pathology: Whole Slide Image Analysis*: A whole slide image is provided. The system runs tumour detection and cell segmentation, returning segmentation results and a summary of the detected region. Success is defined as a completed segmentation run with quantitative output, specifically the detected region and cell count, returned without manual intervention.

3) *Use Case 1 - Interactive DICOM Analysis and Segmentation: Clinical context*. A clinician has a DICOM scan and needs segmentation without knowing which model to use or how to chain the required tools, or simply wants to accelerate their work by dropping the DICOM files and looking directly at the processed and interpreted data.

Setup. The user uploads either a single DICOM volume file or a directory of DICOM slices forming a 3D series through the frontend file selector, and enters the following prompt in the chat to initiate the workflow:

“Analyze this CT scan and run the appropriate segmentation model.”

Workflow. The workflow is illustrated in Figure 2. The agent calls `parse_dicom_file` to extract modality, body part, and acquisition metadata from the DICOM header, then `analyze_image` to determine image type, intensity, and spatial dimensions. Based on these findings, `list_models` returns a filtered list of compatible segmentation models. If the selected model is not cached locally, `download_model` retrieves it from HuggingFace Hub. Inference is then submitted to the background task queue via `run_inference`, which applies a sliding window strategy to process the 3D volume without exceeding GPU memory limits. The agent responds immediately after queuing without waiting for inference to complete. When the background task finishes, an SSE notification is pushed to the frontend and the segmentation results, including per-organ volumes, are displayed in the Results tab.

4) Use Case 2 - Autonomous Pathology ROI Monitoring:

Clinical context. A pathology instrument or PACS system exports region-of-interest (ROI) images to a local directory as

ANALYSIS MODE

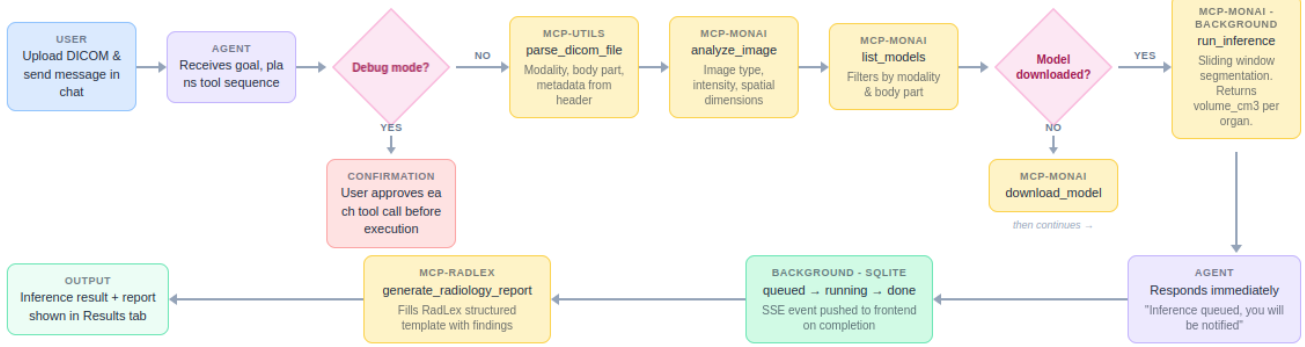


Fig. 2. Use Case 1 - Interactive DICOM analysis and segmentation workflow.

part of its standard output. No human is available to initiate analysis for each arriving file. The goal is for the system to detect each new file, run cell segmentation automatically, and route the result based on a configurable cell count threshold, without any user interaction. Files can also be deposited manually through the Workspaces tab in the frontend, which provides a console showing detection events and agent activity in real time.

Setup. A workspace is registered pointing to the watched directory. The following prompt is configured for that workspace, defining the agent’s behaviour for every arriving file:

“When a new file arrives, check whether it is a pathology ROI image. If it is, run Cellpose cell segmentation and count the cells. If the cell count exceeds 2000, copy the segmentation mask to the output folder and log the result. If the file is not a valid ROI image, move it to the uncertain folder.”

Workflow. The workflow is illustrated in Figure 3. When the polling observer detects a new file, the watcher copies it to the workspace incoming folder and triggers the agent after a three-second debounce window. The agent calls `get_file_metadata` to determine file type, extension, and dimensions, and attempts to classify whether the file is a pathology ROI image. This classification step is an inherent limitation of the current design: reliably determining whether an arbitrary image is a valid ROI would require running a dedicated detection model, effectively doubling the computational cost before any analysis begins. To avoid this overhead, the workspace prompt instructs the agent to make a best-effort classification from file metadata and dimensions alone, and to route uncertain cases to a dedicated folder for manual review rather than proceeding with segmentation. This trades false-negative robustness for efficiency, and is an acknowledged design constraint rather than a failure of the workflow. If the file is classified as a valid ROI image, `segment_cells_2d` is submitted to the background task queue using the Cellpose v4 universal model (cpsam). When the task completes, the result is pushed to the frontend via SSE and displayed in

the Workspaces console. If the cell count exceeds 2000, the segmentation mask is copied to the output folder; otherwise only a JSON log entry is written. Final results, including cell count, mask path, and timestamp, are persisted and accessible in the Results tab.

H. Evaluation Setup

Evaluation was structured in three phases. Phase 1 evaluates both API-based and locally hosted language models for tool-calling reliability in clinical workflows. Phase 2 evaluates whether the agent can operate autonomously in clinical workflows. It first examines how skills provide the workflow constraints that make autonomous execution reliable, showing that the agent follows the correct tool sequence and guard conditions without human intervention. It then compares assisted and autonomous execution of the same task to assess whether human-in-the-loop confirmation changes the outcome, or whether the agent produces equivalent results in both modes. Phase 3 runs the two use cases end to end, comparing autonomous and manual workflows, evaluated by task completion and qualitative assessment of outputs, specifically whether the radiology report was correctly filled and whether segmentation completed successfully.

Language model experiments were run on a laptop with an AMD Ryzen 5 8645HS CPU, 16 GB DDR5 RAM, and an NVIDIA GeForce RTX 3050 6 GB GPU. MONAI inference benchmarks were run on two configurations: the same local machine using CUDA (GPU), and a shared deployment server using CPU-only inference, using MONAI 1.5.1 and PyTorch 2.6.0. Two categories of language model were tested: large API-based models (Gemini 2.0 Flash, Gemini 2.5 Flash, and Gemini 3.0) and locally hosted open-source models of significantly smaller scale (deepseek-r1:7b and llama3.1:8b via Ollama).

The primary imaging dataset used for Use Case 1 evaluation is the **Pancreas-CT** collection from the Cancer Imaging Archive (TCIA) [27], specifically subject PANCREAS_0001. The series consists of **240 DICOM files** totalling **121 MB**, acquired as an abdominal CT volume. All files follow the

WORKSPACE MONITORING

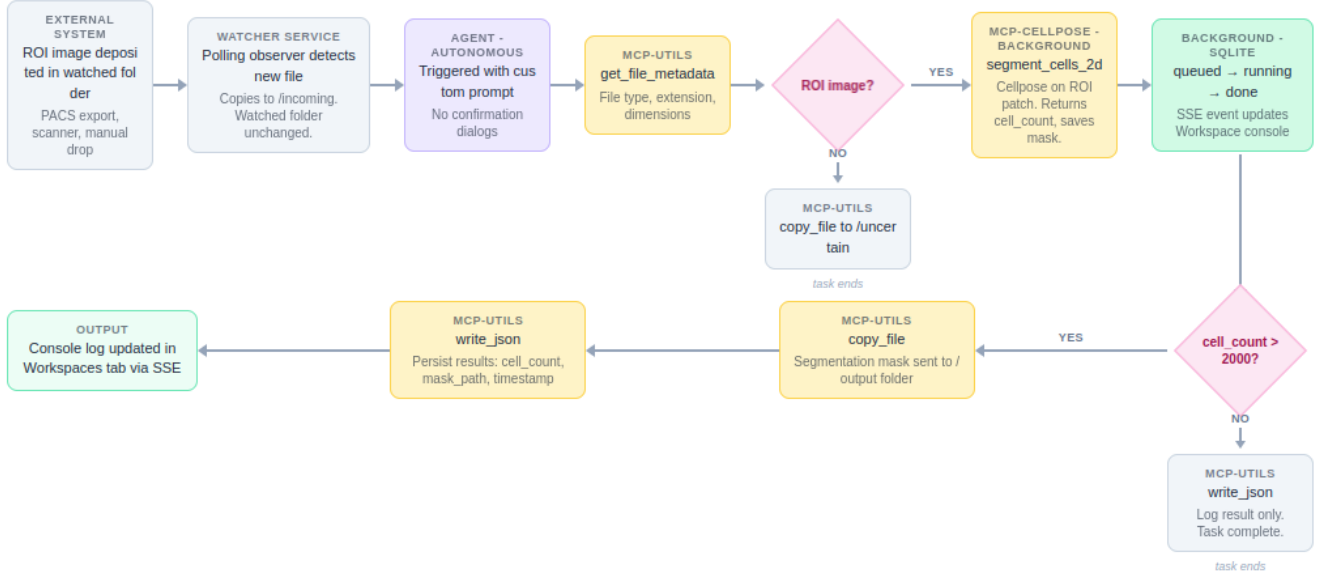


Fig. 3. Use Case 2 - Autonomous pathology ROI monitoring workflow.

DICOM standard and were used without modification. Three MONAI segmentation models were evaluated against this series on both hardware configurations, with results summarised in Table II.

III. RESULTS

A. Phase 1 - Model Experimentation

Five language models were evaluated: three large API-based models (Gemini 2.0 Flash, Gemini 2.5 Flash, and Gemini 3.0) and two locally-hosted open-source models via Ollama (deepseek-r1:7b and llama3.1:8b). The agent loop was capped at 10 iterations per task, meaning a model that cannot reach `goal_achieved` within 10 tool calls fails the task regardless of how many individual tool selections were correct.

1) *Local Models*: Both Ollama models were tested on a laptop with an RTX 3050 Ti GPU at two power profiles: 30W and 60W. At 30W, tool selection was generally correct on the first attempt, and in the majority of cases the first tool called was the appropriate one for the context. However, neither model reliably reached `goal_achieved` within the 10-iteration limit. The models would select correct individual tools but fail to chain them into a complete workflow, running out of iterations before producing a final result. At 60W, outcomes were mixed: some tasks completed successfully within the iteration budget, others did not. A consistent observation across both power profiles was that when a tool call failed, the local models rarely recovered by trying an alternative approach, confirming the need for the repetition guard described in Section II-C.

Each tool call with a locally-hosted model took approximately 20 seconds of inference time, making a 10-iteration

task take up to three minutes of model time alone, before accounting for actual tool execution. This is a fundamental constraint of the model scale rather than a configuration issue.

A key finding from this phase is that two implementation choices significantly influenced local model reliability regardless of power profile: the quality of tool descriptions in each MCP server, and the availability of stateful execution history. Without clear, specific tool descriptions, the models defaulted to plausible-sounding but incorrect tool selections. Without execution history passed back at each iteration, the models repeated the same failing tool calls rather than adapting. Both of these are architectural properties of the system, not model-specific tunings, and both apply equally to the API-based backend.

2) *API-Based Models*: Gemini 2.0 Flash and Gemini 2.5 Flash resolved all of the reliability issues observed with local models. Both models completed multi-step clinical workflows within the iteration budget, correctly sequenced tool calls, and recovered from tool failures by selecting alternatives. Tool call latency was near-instant from the agent's perspective, reducing full task completion time by an order of magnitude compared to local inference. The primary limitations of the API approach are external: API availability is outside the system's control, and all data submitted to the agent is transmitted to a third-party service, which is a concern in privacy-sensitive clinical environments. A further practical consideration is token consumption. Because the agent passes its full execution history back to the LLM at each iteration, token usage grows with every tool call in the loop. Two concrete examples from development illustrate the scale of this problem. In an early version, image bytes were passed directly to the LLM instead

TABLE II
MONAI INFERENCE TIMES: PANCREAS_0001 (240 DICOM FILES, 121 MB). GPU: LOCAL RTX 3050 6 GB (CUDA). CPU (LOCAL): RYZEN 5 8645HS. CPU (SERVER): SHARED DEPLOYMENT SERVER.

Model	Structure	Voxels	Volume (cm ³)	% of Scan	Time (GPU)	Time (CPU, local)	Time (CPU, server)
spleen_ct_segmentation	Spleen	57,284	257.78	0.55%	9 s	27 s	25 s
wholeBody_ct_segmentation	Spleen	43,402	195.31	0.42%	18 s	3 m 56 s	3 m 53 s
	Kidney (right)	30,118	135.53	0.29%			
	Kidney (left)	17,865	80.39	0.17%			
pancreas_ct_dints_segmentation	Pancreas	10,962	49.33	0.11%	27 s	15 m 47 s	14 m 22 s

of a file path, resulting in a single analysis call consuming approximately 60,000 tokens for a 57 MB scan. Separately, the `utils.parse_dicom_directory` tool was including all file paths in its response, adding approximately 27,000 tokens per call for large directories. After fixing both issues, a representative multi-file task begins at roughly 1,720 prompt tokens on the first iteration and grows to approximately 2,352 by the second, reflecting the expected accumulation rate from execution history.

Several measures were introduced to manage this growth. The agent uses Gemini’s native stateful session mode, meaning only the latest tool result is transmitted per turn rather than rebuilding the full history in each prompt. Report generation uses MCP sampling, a lateral LLM call that produces the report narrative without adding to the agent’s context window. The `get_template_schema` tool was removed in a subsequent refactor of the radlex server, eliminating a manual template inspection step that consumed tokens on every report workflow. The skills progressive-disclosure mechanism avoids embedding domain knowledge permanently in the system prompt by loading it only when the agent determines it is needed for the current task. These values and the measures taken represent the state of the system at time of writing; further optimizations remain possible.

3) *Summary*: The results confirm that the system architecture is backend-agnostic and functions with both local and API-hosted models; the gap lies in task completion reliability at current local model scales. Local models at 7–8B parameters can select appropriate tools but cannot reliably complete multi-step workflows within a bounded iteration budget. API-hosted models at significantly larger scale resolve this reliably. The architecture supports both backends without modification, allowing deployment to be matched to the constraints of a given environment: a privacy-critical deployment can use a local model with the understanding that task complexity must be managed, while a deployment where API access is available benefits from reliable autonomous operation. Table III summarises the observed differences.

B. Phase 2 - Autonomous Agent Execution

1) *Skill-Guided Execution*: This experiment evaluates whether loading a skill changes the agent’s tool selection behaviour compared to operating without one, and whether the skill-prescribed sequence is actually followed in autonomous execution.

The **dicom-analysis** skill was used as the test case. Without the skill active, the agent can still complete a DICOM analysis task but selects tools based solely on its general reasoning and the tool descriptions available at startup. With the skill loaded, the agent receives an explicit step-by-step workflow: parse the DICOM header first, call `analyze_image` to confirm modality and body part, call `list_models` filtered to those parameters, download the model if not cached, and only then queue inference. Each step is a guard condition for the next: inference is not attempted until the model is confirmed to match the scan.

In autonomous execution with the skill active, the agent followed this sequence without deviation across all test runs. Guard conditions were respected: on one occasion where `analyze_image` returned an ambiguous modality, the agent called `list_models` with both candidate modalities before selecting, rather than proceeding with the first result. Without the skill, the agent occasionally skipped the analysis step and called `list_models` directly with parameters inferred from the filename, which produced correct results in straightforward cases but would fail on files with non-descriptive names or atypical metadata.

A comparable observation was made in workspace monitoring execution. A deployment trace from March 2026 shows the agent autonomously handling an unrecognised file: it called `utils.move_file`, attempted `monai.analyze_image` (which returned a DICOM header error), created an `uncertain/` subdirectory, moved the file there, and called `goal_achieved`, completing the task in six iterations without any user input. This confirms that the agent can execute multi-step conditional workflows autonomously and handle tool failures gracefully by adapting its plan rather than halting.

The conclusion from this phase is that skills are not required for the agent to complete tasks, but they are what makes autonomous execution *reliable*: they constrain the agent to the correct domain-specific sequence and prevent it from taking plausible but clinically incorrect shortcuts.

A complementary observation concerns the relative efficiency of MCP tools versus skills for the same task. When a direct MCP tool exists for an operation, the agent consistently prefers it over the equivalent skill-based approach. This is expected: the agent reasons over tool descriptions at each iteration and a single tool call is always cheaper than the three-to-four calls required to load a skill (read entry point, optionally read references,

TABLE III
LANGUAGE MODEL COMPARISON FOR AGENTIC TOOL-CALLING

Model	Type	Time/call	Task completion
deepseek-r1:7b (30W)	Local	≈20s	Failed within 10 iter.
llama3.1:8b (30W)	Local	≈20s	Failed within 10 iter.
Both models (60W)	Local	≈20s	Partial
Gemini 2.0 Flash	API	<1s	Consistent
Gemini 2.5 Flash	API	<1s	Consistent

execute). A concrete example from development: extracting DICOM metadata via the `utils.parse_dicom` MCP tool requires one call, whereas the `pydicom` skill achieves the same result through `skills.read_skill_file`, `skills.read_references`, and `skills.execute_script` - three calls and additional context window usage. Skills are therefore most valuable where no direct MCP tool exists, where workflow sequencing across multiple tools must be constrained, or where the domain knowledge to be loaded is too large or too domain-specific to embed permanently in the system prompt.

2) *Assisted vs Autonomous Mode*: The same DICOM analysis task was executed twice: once in assisted mode, where the agent paused before each tool call and presented the proposed tool name and arguments to the user for confirmation, and once in autonomous mode, where all tool calls were executed without interruption. The goal prompt, uploaded files, and active skill were identical in both runs.

The tool call sequences were identical across both executions. The agent selected the same tools in the same order, passed the same arguments, and reached `goal_achieved` in the same number of iterations. No tool call was rejected or modified by the user during the assisted run, and the outputs produced, segmentation results and per-organ volumes in the Results tab, were equivalent.

This result has a direct architectural implication. If assisted mode were a correction mechanism, the two sequences would diverge whenever the user overrode a proposed tool call. The fact that they did not indicates that the agent’s autonomous decisions were already correct, and that the confirmation step added visibility without changing the outcome. Assisted mode is therefore best understood as a transparency and audit mechanism for operators who want to inspect each tool call before it executes, rather than a safety net that catches errors the agent would otherwise make. This in turn validates autonomous mode for event-driven workflows such as Use Case 2, where no user is present: the agent produces the same result whether or not a human is watching.

C. Phase 3 - Use Case Comparison

This phase evaluates the system through two end-to-end clinical workflows covering different imaging modalities and tool combinations. Use Case 1 covers interactive DICOM analysis: a user uploads a CT series, issues a natural language prompt, and the agent autonomously parses the study, selects and runs a segmentation model, and generates a structured

radiology report. Use Case 2 covers event-driven pathology analysis: the workspace monitoring system detects incoming whole-slide image ROIs, triggers the agent without user input, runs cell segmentation via Cellpose, and routes the result based on a threshold rule. Each task was executed under two conditions, with and without a skill loaded, to isolate the effect of domain-specific workflow guidance on agent behavior.

1) *Use Case 1 Results*: Task 1 was initiated with a publicly available DICOM series from the Cancer Imaging Archive and the following prompt:

“Analyze this DICOM series and run the appropriate segmentation model. Use the spleen model and make a report with the findings.”

Task 2 was initiated without a file upload and the following prompt:

“Find the tumor in this lung thumbnail (UID: 2.25.338247016696998394111485786182386609869). Identify and localize any suspicious regions consistent with tumor presence. Within the detected tumor area, estimate and count the number of cells. Then generate a structured report summarizing the findings, including tumor location, size (if inferable), cell count estimate, and any relevant observations or limitations.”

Each task was executed under two conditions. In the skilled condition, the agent loads the corresponding skill file and follows its prescribed sequence as a workflow protocol. In the unskilled condition, the agent selects tools based solely on its general reasoning over the available tool descriptions. Results are summarised in Table IV.

Task 1: DICOM Analysis and Report Generation. Both conditions completed successfully. The agent parsed the DICOM directory, identified the study as an abdominal CT, selected a compatible segmentation model, and submitted inference to the background task queue. Spleen segmentation completed correctly in both runs, yielding a volume of 192.51 cm³. A structured radiology report was produced using a valid RadLex template in each case. The two conditions selected different templates from the catalogue, both appropriate for the study type (Figure ??). First-iteration token counts were nearly identical across conditions (8,204 without skill, 8,251 with skill), consistent with on-demand skill loading. The skilled condition required two additional iterations for the steps prescribed by the workflow protocol (`analyze_image` and `download_model`), increasing total token consumption

TABLE IV
USE CASE 1: EXECUTION METRICS AND OUTCOMES

Condition	Iterations	Total tokens	Time (s)	Completed	Incorrect tool calls
Task 1 without skill	6	80,961	58.3	Yes	None
Task 1 with skill	8	90,973	95.1	Yes	None
Task 2 without skill	8	95,942	117.9	Partial	<code>scale_roi_to_slide</code> called with 200×200 px bbox (30 px max, ignored by agent); <code>radlex.generate_radiology_report</code> applied to pathology specimen
Task 2 with skill	9	135,842	129.6	Yes	None

from 80,961 to 90,973 and wall-clock time from 58.3 s to 95.1 s, with no difference in clinical output.

Task 2: Whole-Slide Image Tumor Detection. Both conditions executed the full cell segmentation sequence: thumbnail retrieval, slide dimension acquisition, ROI coordinate scaling, region extraction, and Cellpose inference. In the unskilled condition, the agent called `ipath.scale_roi_to_slide` with a 200×200 px bounding box, exceeding the 30 px maximum documented in the tool description. The constraint was present in the tool description and available to the agent, but was overridden by the agent’s own spatial estimate. Given the dimensions of the test image (108,000×75,140 px), a 200 px bounding box in thumbnail coordinates maps to a slide region substantially larger than the 2,700×2,700 px retrieval limit of `ipath.fetch_roi`, resulting in a region that was both potentially mislocated and spatially clipped rather than the intended ROI.

Independently, the unskilled condition produced a domain-incorrect report at the final step. The agent called `radlex.generate_radiology_report` on a whole-slide pathology image, yielding a radiology template with fields structured for radiology studies and inapplicable to a tissue specimen (Figure ??). The task is classified as partial: cell segmentation completed but the final report was produced in the wrong clinical domain.

With the **wsi-tumor-detection** skill active, the agent followed the prescribed workflow sequence without deviation. Cell segmentation yielded 428 cells detected in the 2,700×2,700 px ROI. The final report was generated using the pathology template from the **clinical-reports** skill, with fields appropriate to a tissue specimen including specimen information, microscopic description, and diagnosis (Figure ??). The skilled condition consumed 135,842 total tokens against 95,942 in the unskilled condition, reflecting the additional context from loading two skill files, with wall-clock times of 129.6 s and 117.9 s respectively.

Both tasks completed end-to-end without manual intervention, producing clinically structured outputs from raw imaging inputs. Task 1 confirmed that the system correctly orchestrates DICOM parsing, model selection, inference, and report generation autonomously. Task 2 confirmed the same for whole-slide image analysis and cell segmentation, with the additional observation that the skill was necessary to produce a domain-correct report at the final step.

2) *Use Case 2 Results:* Initial testing of the workspace monitoring system used two workspaces configured to validate both the watcher trigger pipeline and the agent’s ability to follow workspace-specific instructions without cross-contamination between workspaces.

The first workspace, **CELLPOSE**, was configured with the prompt:

“Do nothing, just a placeholder for images.”

This prompt was deliberately minimal to test whether the agent would respect a no-op instruction rather than defaulting to unsolicited analysis when a file arrived. Files deposited into this workspace produced no agent action, confirming that the model does not exhibit a bias toward running tools when the prompt does not call for it.

The second workspace, **ROI**, was configured with the following prompt, provided verbatim to demonstrate replicability:

“To this folder there will arrive ROI images, I want you to count the number of cells for each image that arrives and I want that if the number of cells counted is above 100 to send the image to the folder in /app/workspaces/CELLPOSE.”

When `ROI_TEST_IMAGE.jpeg` (2,352×2,673 px, H&E-stained pathology ROI) was deposited into the ROI workspace, the following sequence was observed and is visible in the workspace console (Figure 4): the watcher detected the file within seconds, triggered the agent after the three-second debounce window, and the agent autonomously called `cellpose.segment_cells_2d` without any user interaction. The model reported 1,706 detected cells and, since this exceeded the threshold of 100, moved the image to `/app/workspaces/CELLPOSE` as instructed. The full sequence from file drop to analysis completion took approximately 30 seconds on a local GPU (RTX 3050).

This test confirmed end-to-end operation of the watcher service, the debounce trigger, the agent’s tool selection, and cross-workspace file routing via a plain-language threshold rule.

3) *Inference Hardware Requirements:* Inference performance is tightly coupled to available hardware, and the gap between CPU and GPU execution has direct clinical implications. Table II summarises measured times across GPU and CPU configurations for both the local development machine and the shared deployment server.

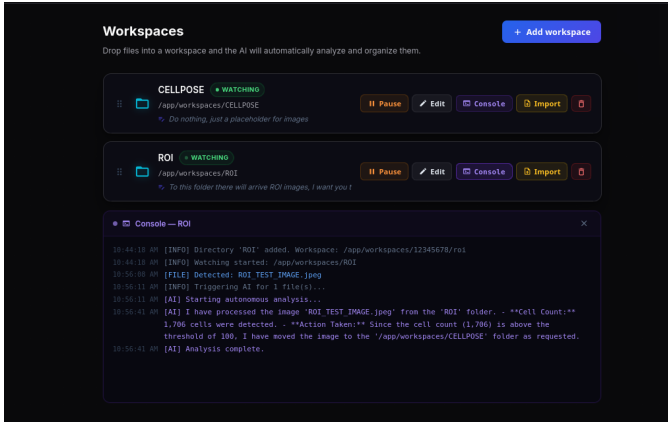


Fig. 4. Workspace console showing autonomous detection and routing: ROI_TEST_IMAGE.jpeg detected, 1,706 cells counted, image moved to CELLPOSE workspace.

The Cellpose v4 model (cpsam) is a SAM-based architecture designed for GPU execution. On CPU, inference took 95 seconds on a 64×64 pixel test image. A real pathology ROI image at typical resolution ($2,352 \times 2,673$ px) completed in **12 minutes 37 seconds** on a local Ryzen 5 8645HS CPU, and ran for approximately three hours on the deployment server before hanging. The contrast with the MONAI inference times in Table II is notable: MONAI times are nearly identical across both machines because the server is configured with `OMP_NUM_THREADS=1` and `MKL_NUM_THREADS=1`, pinning it to a single thread where per-core speed is comparable. Cellpose on CPU is unrestricted and scales with available cores; the deployment server runs as a virtual machine with only 4 cores allocated from a Ryzen 9 5950X host, introducing both core-count and VM overhead constraints that explain the order-of-magnitude difference relative to the local machine. On the local machine with an RTX 3050 GPU (CUDA), the same image completed in **30 seconds**. On the deployment server with an RTX 3060 GPU (CUDA 12.1), the same image completed in **28 seconds**, confirming that GPU execution eliminates the VM CPU bottleneck entirely and yields equivalent performance across both hardware configurations. GPU execution is therefore not an optional accelerator but a practical prerequisite for the pathology use case [28]. The MONAI inference pipeline is similarly GPU-dependent: on the test machine (RTX 3050 6 GB), certain models exceeded available VRAM and caused the subprocess to terminate without producing any output or error message. This failure mode was only diagnosed after adding stderr capture to the task runner, which then surfaced the underlying PyTorch out-of-memory exception. These findings shaped deployment requirements: the system is shipped with a GPU-enabled Docker configuration as the recommended deployment target, with CPU fallback available for development and testing only.

A practical advantage of both workflows, independent of accuracy, is the convenience the system provides. Processing speed depends on available hardware, cached models,

and firmware, and is therefore highly deployment-specific. What the architecture guarantees regardless of hardware is that long-running operations are queued automatically, results are persisted, and the clinician can return to completed outputs without having been present during execution. In the workspace monitoring scenario this translates directly to overnight or unattended processing: files deposited at the end of a working day are analyzed and results are available the next morning. Meaningful evaluation of this property at scale, however, requires a broader set of test datasets and input types covering diverse modalities, resolutions, and acquisition protocols, which represents the primary direction for further experimental work.

IV. DISCUSSION

A. MCP as a Practical Integration Layer

The Background identifies fragmentation as a structural problem in clinical AI: tools exist, but no standard layer connects them, forcing each institution to build and maintain custom integrations. MCP was evaluated in this work as a candidate solution to that problem. The results support that framing. Six servers covering distinct clinical capabilities were integrated without any agent-side connector code: each server exposes its tools through the MCP protocol, and adding a new server required no changes to the agent or any existing component. More importantly, this composability held under real workflow conditions: both use cases required the agent to sequence operations across multiple servers dynamically, and in both cases it did so correctly without any hardcoded knowledge of how those servers were implemented.

The broader implication is that MCP could reduce the integration overhead that currently slows AI adoption in clinical settings. Under the conventional approach, connecting a new tool to an existing workflow requires custom development work regardless of how well-designed the tool itself is. Under an MCP-based architecture, the integration contract is defined by the protocol: a conforming server is immediately accessible to any conforming agent. If this holds as more clinical tools adopt MCP, the bottleneck shifts from integration engineering to tool capability itself, which is a more tractable problem. Singh et al. argue this conceptually [21]; the use cases presented here provide evidence that it holds in practice.

One finding qualifies this picture. MCP decouples the agent from a server's implementation, but it does not decouple the agent's decision-making from the quality of each tool's natural language description. Vague or ambiguous descriptions caused incorrect tool selections even when the correct tool was available, an effect observed consistently with local models and to a lesser degree with API-based models. This means the integration problem is not fully solved by adopting MCP: it is partially displaced from adapter code to description authorship. That is a smaller and more localised problem, but it is not zero. A further mitigation observed in this system is that skills partially compensate for this: when a skill is active, the agent follows an explicit protocol that names specific tools, reducing its dependence on description quality for sequencing

decisions [24]. MCP and skills are therefore complementary: MCP provides the integration layer, skills provide the domain knowledge that makes reliable use of that layer possible. Neither alone is sufficient for complex clinical workflows.

One open question is how this composability scales beyond the six servers evaluated here. The architecture imposes no limit on the number of registered servers, and adding a new one requires no changes to the agent. However, all tool descriptions are loaded into context at startup, and as the catalogue grows the agent’s ability to select the correct tool from an increasingly large set is expected to degrade, a direct instance of the failure mode where options buried deeper in a long context are more likely to be overlooked or misapplied [26]. Skills that function as workflow guides partially mitigate this by prescribing explicit tool sequences, reducing the risk of the agent selecting the wrong tool from an increasingly large catalogue, but they do not reduce the size of the tool list itself. A natural extension would be a multi-agent architecture in which specialist agents each expose only the tools relevant to their domain, keeping individual context windows bounded regardless of how many total servers exist. This design was not evaluated here and remains future work.

B. Verified Autonomous Execution

The finding that assisted and autonomous mode produced identical tool sequences has a direct implication for how human oversight of agentic clinical AI should be understood. If the confirmation step changed outcomes, assisted mode would be a safety mechanism: the human would be catching and correcting errors the agent would otherwise make. The fact that it did not indicates that the agent’s decisions were already correct, and that the confirmation step added visibility without adding control. Assisted mode is therefore best understood as a transparency and audit mechanism for operators who want to inspect each tool call before it executes, rather than a safety net that catches errors the agent would otherwise make. This in turn validates autonomous mode for event-driven workflows such as Use Case 2, where no user is present: the agent produces the same result whether or not a human is watching.

C. Skills as Workflows: When Guidance Changes Behaviour

The two use cases clarify when a skill operating as a workflow changes the agent’s behaviour and when it does not. In Use Case 1, the skill made no difference to the outcome: the DICOM analysis workflow presents no ambiguous decision forks, so the agent reached the correct sequence through general reasoning alone. The skill enforces protocol adherence on top of an already correct execution, which matters for reproducibility but does not change the result. Use Case 2 produced a different outcome, with two clinically incorrect decisions in the unskilled condition (Section III).

The two failures illustrate distinct mechanisms by which a workflow skill corrects agent behaviour. The first is a *constraint enforcement failure*: the tool description encoded the correct limit, but the agent did not apply it when its own spatial estimate exceeded it. The skill resolved this by making

the constraint an explicit instruction in the workflow rather than a property of the tool schema. The second is a *domain selection failure*: two tools were available that could both produce a structured report, and the agent selected the tool with greater salience in its context window. The skill resolved this by naming the correct tool explicitly and ruling out the incorrect one, a decision the agent could not make reliably from tool names and descriptions alone.

The practical implication is that skills as workflows are most valuable at decision forks where a plausible but incorrect option exists: either a constraint the agent would not enforce unprompted, or a choice between tools where domain appropriateness cannot be inferred from names and descriptions alone. Where no such fork exists, as in Use Case 1, the skill adds no behavioural change and its value is limited to reproducibility guarantees.

D. Clinical Accuracy and Validation Scope

The evaluation in this work measures task completion: whether the agent correctly orchestrated the required tools, followed the prescribed workflow, and produced an output. It does not measure clinical correctness: whether the segmentation outputs are accurate enough to inform a clinical decision. These are distinct properties, and conflating them would overstate what the system has been shown to do.

Segmentation accuracy for the models used (spleen, pancreas, and whole-body CT segmentation via MONAI) has been established in their respective publications and benchmarks. What has not been evaluated here is whether the agent’s orchestration introduces errors that would not arise in a direct model invocation: whether the correct model is consistently selected for a given scan, whether the inference parameters passed by the agent are appropriate, and whether the structured report correctly maps segmentation findings to the right template fields. These are failure modes specific to the agentic layer, not the underlying models, and they require dedicated clinical validation to assess.

A rigorous evaluation would require comparison of agent-produced outputs against ground-truth annotations by trained radiologists, Dice similarity scores across a representative dataset of scan types and qualities, and prospective testing across institutions with different equipment and acquisition protocols. Until that validation is done, the system should be understood as a workflow automation tool that reduces the manual effort of orchestrating clinical imaging pipelines, not as a clinically validated decision support system. That distinction is not a limitation unique to this work, as it applies to any system that introduces an agentic orchestration layer over existing models, but it needs to be stated explicitly.

E. Privacy vs Reliability in Clinical Deployment

The results expose a tension that has no clean resolution in the current system. While API-based models completed multi-step clinical workflows reliably, locally hosted models at 7–8B parameters did not. The practical consequence is that reliable autonomous operation currently entails transmitting clinical

metadata, including patient identifiers present in DICOM headers, to a third-party cloud service, which is a significant concern in clinical environments governed by data protection regulations such as GDPR and HIPAA. The two properties that matter most for clinical adoption, privacy and reliability, pull in opposite directions under current model constraints.

This constraint reflects the computational resources available for this evaluation rather than the state of local model capability in general. Larger open-weight models capable of more reliable multi-step reasoning exist but require GPU resources beyond what was available in this work, and beyond what most clinical deployments can provide without dedicated AI infrastructure. The models evaluated locally represent what is practically deployable on standard institutional hardware, which is the relevant constraint for widespread clinical adoption. The architecture supports both backends without modification, which means the system can be deployed in privacy-critical environments using a local model. The limitation observed here is not inherent to local inference but specific to the 7–8B parameter models tested: a sufficiently capable open-weight model running locally would satisfy both requirements simultaneously. As hardware accessible to clinical institutions improves and larger open-weight models become practically deployable, the same architecture would benefit without any changes to the orchestration layer.

Several paths could close this gap. Fine-tuning a smaller model specifically on tool-calling sequences in clinical workflows could improve reliability without requiring frontier-scale parameters. On-premise deployment of larger models, where the hardware investment is justified by the privacy requirement, is a viable option for well-resourced institutions. A hybrid approach, where a local model handles simple routing decisions and an API model is invoked only for complex multi-step reasoning on de-identified data, could balance both concerns. None of these were evaluated in this work, and they represent concrete directions for future investigation [12], [16], [20].

V. CONCLUSIONS

This work set out to demonstrate that an agentic AI system using MCP as a standardized interoperability layer can autonomously orchestrate multi-step clinical imaging workflows without custom integration code. The results support that claim. Both use cases completed end-to-end without manual intervention, six MCP servers were integrated without agent-side connector code, and the skill-based governance mechanism demonstrably changed agent behavior in the pathology task, preventing domain-incorrect tool selection that occurred without it. The finding that assisted and autonomous modes produced identical tool sequences confirms that the agent’s decisions were correct independently of human oversight, validating unattended operation for event-driven workflows.

The comparison between API-hosted and locally-hosted models exposed a tension that clinical deployment cannot currently avoid. Reliable multi-step tool-calling requires model scale that exceeds what most institutions can run locally,

yet transmitting clinical data to a third-party API raises data protection concerns under GDPR and HIPAA. This is not a limitation unique to this system; it reflects the current state of open-weight model capability relative to the computational resources available in clinical environments [20]. A hybrid approach, where a local model handles routine decisions on identified data and an API model is invoked only for complex reasoning on de-identified inputs, represents a concrete direction for resolving this tension.

Logging and auditing, implemented here primarily as a debugging and transparency mechanism, proved to be more valuable than anticipated. The workspace monitoring console provided a step-by-step trace of every autonomous agent action, which was essential both for diagnosing failure modes and for building confidence in unattended operation. As agentic systems take on more consequential clinical tasks, this kind of decision traceability will become a compliance requirement rather than a convenience [12], [16]. Security considerations, including tool-level access control and audit trail integrity, were not the focus of this work but represent a necessary area of development before any production clinical deployment [23].

Several directions remain for future work. Token consumption optimization is the most immediate: further reducing context window growth through smarter execution history pruning and selective skill loading would lower API costs and extend the effective iteration budget. Prompt engineering and fine-tuning smaller models specifically on clinical tool-calling sequences could close the reliability gap between local and API backends without requiring frontier-scale compute. The current background task queue serializes inference for safety but is not designed for multi-tenant scale; a dedicated worker process per GPU using a proper task queue would be the right architecture for production deployment across multiple concurrent users. Finally, and most importantly, the evaluation presented here measures workflow orchestration correctness, not clinical accuracy. A rigorous validation requires Dice score comparisons against ground-truth annotations across a representative dataset, prospective testing across institutions with different equipment, and review by trained clinicians. That validation is the necessary next step before MedGov-AI can be considered for any real clinical use.

The broader contribution of this work is a demonstration that MCP, combined with a skills system for domain-specific governance, provides a practical and extensible integration layer for clinical AI. The approach is not limited to the servers implemented here: any clinical tool that can be wrapped in an MCP server becomes immediately accessible to the agent without changes to the orchestration layer. As the clinical AI ecosystem matures and more tools adopt MCP, the integration overhead that currently limits adoption [14] shifts from adapter engineering to tool capability itself, which is a more tractable problem for the field to solve.

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